

Lab Notebook

Charlie Fioriglio

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Photodetachment Code Suite

My aim is to make a software package for calculating photodetachment total and differential cross sections. This work will expand on the existing open source software `ezDyson` [?] [?] with customizable averaging, wavefunction and continuum visualizations, and advanced continuum functions options. Specifically, this code will allow the treatment of electron-molecule interactions in the continuum through both a point dipole potential [?] and physical dipole potential [?] [?]. Both methods will be expanded on below.

1 Dyson Orbital Reader

The first necessary thing to create is a Dyson orbital reader. The code must be able to read Q-Chem Dyson Orbital output files and build the DO from an Atomic Orbital basis.

1.1 Q-Chem Output Parsing

The Dyson Orbital reader begins by parsing the Q-Chem output file to extract three key components: molecular geometry, basis set definitions, and Dyson orbital coefficients.

1.1.1 Geometry Extraction

The code searches for the `Standard Nuclear Orientation` block in the Q-Chem output. It reads the atomic symbols and their Cartesian coordinates (in Angstroms), converting them to Bohr radii for internal calculations.

1.1.2 Basis Set Parsing

The basis set information varies depending on the atom and shell type. The parser extracts the `Basis Set` block, identifying each atom and its associated shells (S, P, D, F, G, etc.).

1. **Atom Identification:** The parser links basis blocks to specific atoms in the geometry.
2. **Shell Processing:** For each shell, the code reads the number of primitives, their exponents, and contraction coefficients.
3. **Pure vs. Cartesian Detection:** The parser checks the `purecart` setting in the output to determine if shells with $\ell \geq 2$ should be treated as pure spherical harmonics or Cartesian functions. This is critical for correctly enumerating the basis functions (e.g., 5 vs. 6 functions for D-shells).

1.1.3 Dyson Coefficient Extraction

The Dyson orbitals are defined as linear combinations of the basis functions. The parser locates the `Decomposition over AOs` sections. For each Dyson orbital (corresponding to a specific ionization transition), it reads the coefficients C_μ that weigh each basis function contribution. The code handles both "Left" and "Right" Dyson orbitals if biorthogonal sets are used, though typically the "Right" orbital is primary.

1.2 Dyson Orbital Construction Details

1.2.1 Primitives and Basis Functions

The fundamental building blocks of the Dyson orbital are Cartesian Gaussian primitive functions. A primitive function centered at $\mathbf{R} = (x_0, y_0, z_0)$ with exponents l_x, l_y, l_z and decay constant α is defined as:

$$g_{ijk}(\mathbf{r}; \alpha, \mathbf{R}) = N(l_x, l_y, l_z, \alpha) \cdot (x - x_0)^{l_x} (y - y_0)^{l_y} (z - z_0)^{l_z} \exp(-\alpha |\mathbf{r} - \mathbf{R}|^2) \quad (1)$$

The normalization constant N ensures that the self-overlap of the primitive is unity. For a Cartesian Gaussian, it is given by:

$$N(l_x, l_y, l_z, \alpha) = \left(\frac{2\alpha}{\pi}\right)^{3/4} \left(\frac{(4\alpha)^\ell}{(2l_x - 1)!!(2l_y - 1)!!(2l_z - 1)!!}\right)^{1/2} \quad (2)$$

where $\ell = l_x + l_y + l_z$ is the total angular momentum quantum number, and $n!!$ denotes the double factorial.

Contracted basis functions (Atomic Orbitals, AOs) are formed as linear combinations of these primitives:

$$\phi_{AO}(\mathbf{r}) = N_{contraction} \sum_p c_p g_p(\mathbf{r}) \quad (3)$$

The contraction coefficients c_p are read from the basis set definition. The code computes the overlap matrix $S_{pq} = \langle g_p | g_q \rangle$ of the primitives numerically or analytically, and then determines $N_{contraction}$ such that $\langle \phi_{AO} | \phi_{AO} \rangle = 1$. Specifically:

$$N_{contraction} = \frac{1}{\sqrt{\sum_{p,q} c_p S_{pq} c_q}} \quad (4)$$

1.2.2 Pure vs. Cartesian Functions and Ordering

For shells with angular momentum $\ell \geq 2$, functions can be expressed in either Cartesian or Spherical Harmonic ("pure") representations.

Cartesian Shells Cartesian shells contain all combinatoric possibilities of (l_x, l_y, l_z) summing to ℓ . The code enumerates these in the order of nested loops: l_x from ℓ down to 0, and l_y from $\ell - l_x$ down to 0.

Pure Shells Pure shells correspond to the $2\ell + 1$ spherical harmonic functions. These are constructed as specific linear combinations of Cartesian primitives. The coefficients used in the Python code match the Q-Chem ordering (AOorder=1) derived from the reference C++ implementation.

D-Shells ($\ell = 2$):

$$\begin{aligned} d_{xy} &= (1.0) \cdot xy \\ d_{yz} &= (1.0) \cdot yz \\ d_{z^2} &= (1.0) \cdot z^2 - 0.5 \cdot x^2 - 0.5 \cdot y^2 \\ d_{xz} &= (1.0) \cdot xz \\ d_{x^2-y^2} &= \left(\frac{\sqrt{3}}{2}\right) \cdot x^2 - \left(\frac{\sqrt{3}}{2}\right) \cdot y^2 \end{aligned}$$

F-Shells ($\ell = 3$):

$$\begin{aligned}
f_1 &= 3\sqrt{\frac{5}{8}}x^2y - \sqrt{\frac{5}{8}}y^3 \\
f_2 &= xyz \\
f_3 &= 4\sqrt{\frac{3}{8}}yz^2 - \sqrt{\frac{3}{8}}y^3 - \sqrt{\frac{3}{8}}x^2y \\
f_4 &= z^3 - 1.5x^2z - 1.5y^2z \\
f_5 &= 4\sqrt{\frac{3}{8}}xz^2 - \sqrt{\frac{3}{8}}x^3 - \sqrt{\frac{3}{8}}xy^2 \\
f_6 &= \frac{\sqrt{15}}{2}x^2z - \frac{\sqrt{15}}{2}y^2z \\
f_7 &= \sqrt{\frac{5}{8}}x^3 - 3\sqrt{\frac{5}{8}}xy^2
\end{aligned}$$

G-Shells ($\ell = 4$):

$$\begin{aligned}
g_1 &= \frac{\sqrt{35}}{2}x^3y - \frac{\sqrt{35}}{2}xy^3 \\
g_2 &= 3\sqrt{\frac{35}{8}}x^2yz - \sqrt{\frac{35}{8}}y^3z \\
g_3 &= 3\sqrt{5}xyz^2 - \frac{\sqrt{5}}{2}x^3y - \frac{\sqrt{5}}{2}xy^3 \\
g_4 &= 4\sqrt{\frac{5}{8}}yz^3 - 3\sqrt{\frac{5}{8}}x^2yz - 3\sqrt{\frac{5}{8}}y^3z \\
g_5 &= z^4 + \frac{3}{8}x^4 + \frac{3}{8}y^4 - 3x^2z^2 - 3y^2z^2 + 0.75x^2y^2 \\
g_6 &= 4\sqrt{\frac{5}{8}}xz^3 - 3\sqrt{\frac{5}{8}}x^3z - 3\sqrt{\frac{5}{8}}xy^2z \\
g_7 &= 1.5\sqrt{5}x^2z^2 - 0.25\sqrt{5}x^4 - 1.5\sqrt{5}y^2z^2 + 0.25\sqrt{5}y^4 \\
g_8 &= \sqrt{\frac{35}{8}}x^3z - 3\sqrt{\frac{35}{8}}xy^2z \\
g_9 &= \frac{\sqrt{35}}{8}x^4 + \frac{\sqrt{35}}{8}y^4 - \frac{3\sqrt{35}}{4}x^2y^2
\end{aligned}$$

1.2.3 Dyson Orbital Assembly

The Molecular Orbital (MO) or Dyson Orbital (DO) is constructed by summing over all basis functions μ :

$$\Psi_{DO}(\mathbf{r}) = \sum_{\mu} C_{\mu} \phi_{\mu}(\mathbf{r}) \quad (5)$$

The code iterates through atoms and shells, generating the appropriate Cartesian or Pure functions for each shell, and applies the corresponding MO coefficients C_{μ} read from the Q-Chem output.

Once built on a 3D grid, the total norm is checked numerically:

$$\text{Norm} = \sum_{ijk} |\Psi(x_i, y_j, z_k)|^2 dV \quad (6)$$

If the norm deviates significantly from 1, the wavefunction is renormalized.

Finally, the orbital centroid is calculated to center the grid for visualization or further processing:

$$\mathbf{R}_{cent} = \frac{\langle \Psi | \mathbf{r} | \Psi \rangle}{\langle \Psi | \Psi \rangle} \quad (7)$$

The grid coordinates are then shifted such that $\mathbf{R}_{cent}$ is at the origin if recentering is enabled.

1.3 Visualization

The visualization module generates an isosurface of the 3D Dyson Orbital. It creates a uniform grid spanning the molecular extent, evaluates wavefunction values $\Psi(\mathbf{r})$ at each grid point, and likely exports this data for plotting (e.g., using Plotly or Matplotlib isosurfaces). The positive and negative phases of the wavefunction are typically colored differently (e.g., red/blue) to indicate phase information.

2 Cross Section Calculations

Total photodetachment cross sections measure the probability of electron detachment for a particular initial and final molecule (reflected in the Dyson orbital) and a given photon energy. The gross section for a detachment angle α is given by

$$\sigma[\alpha] = \frac{8\pi k E}{c} \sum_{klm} |C_{klm}[\alpha]|^2 \quad (8)$$

where E is the photon energy, c is the speed of light, k is the electron momentum

$$k = \sqrt{2m_e e K E} \quad (9)$$

and

$$C_{klm}[\alpha] = i^l \int \phi^d(\mathbf{r}) r_\alpha \psi_{klm}^{el} d\mathbf{r} \quad (10)$$

ψ_{klm}^{el} is the l, m component of the plane wave expansion of the continuum function.

2.1 Transition Operator

r_α is the α component of the transition dipole operator μ .

$$r_\alpha = \hat{\alpha} \cdot \vec{r}_{el} \quad (11)$$

For the total cross section, only $\alpha = x, y, z$ need to be considered and averaged over.

2.2 The Continuum

2.2.1 Plane Wave Expansion

$$\psi_{klm}^{el} = Y_{lm}(\hat{r}) R_l(kr) \quad (12)$$

what R_l is a spherical Bessel function to order l .

2.3 Calculating Values

To execute the sum in (8) you will need to go over x, y , and z polarizations, from 0 to l_{max} (5 should be sufficient) for each polarization, and from $-l$ to l for each l . Execute each integral and add them for a k . All values are in atomic units. Instead of $|C_{klm}|^2$, we used the more accurate $(C_{klm}^L)^* C_{klm}^R$ where L and R reference the Dyson orbitals used.

For the final sigma values divide by three to account for the averaging over xyz and multiply by the Dyson orbital norms $norm_L \times norm_R$. Spin degeneracy will also need to be accounted for, this will introduce a factor of two, if applicable.

2.4 Relative Cross Sections

For relative cross sections, one will need the vibrational transitions and FCFs of the neutral. The relative cross sections now have k_v where v is the vibrational channel, and $\sigma_{rel_v} = FCF_v^2 \frac{\sigma_v}{\sum_{v'=0}^{v'=v_{max}} \sigma_{v'}}$